

A deep neural network model with spectral correlation function for electrocardiogram classification and diagnosis of atrial fibrillation

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ABSTRACT

Atrial Fibrillation (AF) is a common type of irregular heartbeat, and early detection can significantly improve treatment outcomes and prognoses. Single-lead Electrocardiogram (ECG) devices are under extensive scrutiny for monitoring patients' heart health worldwide. Standardized ECG signal monitoring has demonstrated a significant reduction in mortality rates associated with severe cardiovascular diseases. However, the automatic detection method for AF requires significant improvement. This study presents a novel approach that utilizes the cyclostationary analysis of ECG signals, uncovering a spectral hidden periodicity between the QRS-T (the main wave components representing electrical activity in the heart) complexes of the ECG signal through the Spectral Correlation Function (SCF). To validate the proposed method's performance, the single ECG's SCF coefficients are applied to the Convolutional Recurrent Neural Network (CRNN), which consists of convolutional and long short-term memory (LSTM) layers, on the 2017 PhysioNet challenge dataset. The obtained results demonstrate that the proposed approach efficiently represents ECG signals through SCF coefficients, leading to the accurate detection of AF with an average accuracy of 92.76% and an average F1-score of 89.1%.

1. Introduction

1.1. Background

Atrial Fibrillation (AF) is a prevalent cardiovascular disease that presents significant challenges for patients, physicians, and healthcare systems [1]. AF is characterized by an irregular and abnormal heart rate [2]. Enhancing computer-aided diagnostic systems for AF can improve physicians' ability to detect it, reduce the burden on healthcare systems, and minimize the risk of irreversible side effects for patients with AF.

During the last decade, lots of researchers allocate their studies on Electrocardiogram (ECG) signal analyzing and developing algorithms for AF detecting [3]. The availability of large open-access ECG databases on the internet and the organization of reputable competitions like PhysioNet challenges have greatly facilitated researchers in developing new machine learning techniques, including Deep Learning (DL) approaches. As of now, methods for AF detection can be categorized into two main groups: (i) methods based on hand-crafted features, and (ii) approaches based on DL.

Hand-crafted features are typically obtained by extracting various features from the ECG rhythm, such as R-peaks (RR) intervals [4,5], ECG morphology, such as the presence or absence of f-waves and P-waves [6,

7], or both [8,9]. Certain studies have explored the fusion of DL with hand-crafted features by utilizing a feature vector that incorporates diverse features as input to the DL network [10,11]. This approach aims to enhance the interpretability of the DL model and potentially improve the overall performance of the combined system [12]. Alternatively, some studies have utilized specific ECG information, such as the RR interval signal, as input to a DL based algorithm. This approach can expedite the learning process of the network; however, the reliability of these algorithms is heavily contingent on the accuracy of identifying the R-peak, and they may prove less effective for ECG leads with double or inverted R-peaks [13]. Despite these potential advantages, extracting hand-crafted features for this approach poses challenges and demands a comprehensive understanding of the nature of ECG signals.

Certain DL based studies have incorporated attentional methods inspired by human vision mechanisms, enabling models to swiftly identify task-relevant regions and extract critical information from input data. For instance, in Ref. [14], a residual network with a temporal attention block was developed specifically for AF detection. Mäkynen et al. [15] further advanced this by employing a compressed DL network for AF diagnosis, while Li et al. [16] introduced a self-complementary attentional convolutional neural network. To address recurrent neural network (RNN) limitations, Islam et al. [17] proposed a dual-structured

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hierarchical attention-based RNN model.

Beyond these advancements in attention-based networks, various DL approaches have been proposed for AF detection. Some studies have employed a one-dimensional convolutional neural network (CNN) directly on single-lead ECG data [18]. Others have broadened the perceptual scope of the model by transforming ECG data into two-dimensional representations through methods such as the short-time Fourier transform [19,20], chirplet transform [21] and wavelet transform [2,22], thereby enhancing the DL model's capacity. Additionally, hybrid approaches combining CNN and RNN layers have been used to leverage both spatial and temporal features in ECG signals [23,24]. Study [25] further improved data visualization by employing the Poincaré diagram, which aids in distinguishing AF patterns.

While there have been some commendable studies conducted in the area of ECG analysis, a gap in research pertaining to the investigation of cyclostationary correlation between QRS-T complexes of ECG signals still exists. This study proposes a novel method that utilizes Spectral Correlation Function (SCF) to identify micro-repetitive patterns within QRS-T complexes of ECG signals. The method estimates cyclic correlation connections and leverages deep learning techniques to recognize AF by utilizing all available cyclic correlation information.

The structure of the paper is as follows: The section titled "Literature Review on Spectral Correlation Function" summarizes key studies on SCF. Following this, "Theoretical Background" provides an in-depth explanation of SCF concepts. The "Materials and Methods" section details the dataset and methodology used in this study. The "Results and Discussion" section presents the findings, offering a comprehensive discussion of the performance assessment. Finally, the "Conclusion" summarizes the main insights and provides an overview of the study's contributions.

2. Literature review on spectral correlation function

Certain physiological phenomena, such as brain and heart activity, exhibit complex statistical behaviors that remain challenging to define precisely. One such behavior is cyclostationarity, an intuitive yet not fully proven concept suggesting hidden periodicities within the data. For instance, ECG signals are hypothesized to exhibit cyclostationary characteristics, indicating underlying rhythmic patterns. This study investigates this hypothesis, building on previous research into biomedical signals [26,27]. Cyclostationary signals refer to regular variations in the statistical properties of the signal over time [28]. The SCF is a spectral statistical representation of a signal, obtained by estimating the cross-spectral density of the signal and its frequency-shifted version, which provides richer information compared to conventional power spectral density. Cyclostationary analysis has been applied across various fields, including biomedical signal processing, to assess its efficacy. For example, Panahi et al. utilized cyclostationarity for brain-computer interference analysis [29], while Mihanidoost et al. explored SCF in heart analysis and epilepsy diagnosis [26,30], respectively. In Ref. [26], however, SCF was applied to a single QRS-T complex in ECG signals, representing SCF coefficients with limited features, which led to significant loss of cyclic correlation and missed valuable patterns across multiple QRS-T complexes—information crucial for analyzing RR intervals and other repetitive patterns in ECG. Our findings suggest that applying SCF to the entire ECG signal, encompassing multiple QRS-T complexes, could offer meaningful insights for atrial fibrillation diagnosis.

3. Theoretical background

In this section, the fundamental principles of the suggested approach will be elaborated upon. Gardner was the first to introduce the principal theory of cyclostationary signals and present different algorithms for estimating the one-dimensional SCF [28]. Subsequently, two time-smoothing algorithms based on Fast Fourier Transform (FFT),

namely the FFT Accumulation Method (FAM) and the Strip Spectral Correlation Analyzer were developed and discussed in Ref. [31]. For a discrete-time signal $z(n)$ of length N , the cyclic autocorrelation function can be defined as follows [28]:

$$R_z^\alpha(k) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N [z(n+k)e^{-j\pi\alpha(n+k)}] [z(n)e^{-j\pi\alpha n}]^* \quad (1)$$

k represents the delay between observations of the signal $z(n)$ and α denotes the cyclic frequency, which is dependent on the degree of frequency shift. By applying Fourier transform on the $R_z^\alpha(k)$, the SCF is achieved, as:

$$S_z^\alpha(f) = \sum_{k=-\infty}^{+\infty} R_z^\alpha(k) e^{-j2\pi f k} \quad (2)$$

where f is the cross spectrum frequency variable. If the signal has finite average power, the numerical value of cyclic frequency is countable. Additionally, when $\alpha = 0$, the ordinary power spectral density can be represented by the following equation:

$$S_z(f) = S_z^{\alpha=0}(f) = \sum_{k=-\infty}^{+\infty} R_z^0(k) e^{-j2\pi f k} \quad (3)$$

It is clear that the SCF offers a greater range of information compared to the standard power spectral density. In the FAM algorithm, the cyclic cross-periodogram is represented as:

$$S_{zT}^{\alpha_i}(n, f)_{\Delta t} = \frac{1}{T} \langle Z_T(n, f + \frac{\alpha_i}{2}) Z_T^*(n, f - \frac{\alpha_i}{2}) \rangle_{\Delta t} \quad (4)$$

Equation (4) explains how the frequency components of a signal are correlated over a specific time interval. $Z_T(n, f + \frac{\alpha_i}{2})$ and $Z_T^*(n, f - \frac{\alpha_i}{2})$ represent the complex demodulation of the captured $z(n)$ by a data capturing window of length T , which can slide over $z(n)$ over the time span of Δt [31]. Therefore, using the correlation of complex demodulations over a duration of Δt seconds, the cyclic cross periodogram for a particular α_i and f_j can be computed as follows:

$$S_{zT}^{\alpha_i}(n, f_j)_{\Delta t} = \sum_r Z_T(n, f_j + \frac{\alpha_i}{2}) Z_T^*(n, f_j - \frac{\alpha_i}{2}) g(n-r) \quad (5)$$

where $g(n)$ is a data capturing window with $\Delta t = NT_s$ width. The FAM algorithm employs Fourier transform to smoothen the time domain. As the frequency changes from α_i to $\alpha_i + \varepsilon$, the algorithm's output is represented as [31]:

$$S_{zT}^{\alpha_i+\varepsilon}(n, f_j)_{\Delta t} = \sum_r Z_T(n, f_j + \frac{\alpha_i}{2}) Z_T^*(n, f_j - \frac{\alpha_i}{2}) g(n-r) e^{-j2\pi\varepsilon r T_s} \quad (6)$$

Equation (6) can be modified by discretizing the value of ε to be $q\Delta\alpha$:

$$S_{zT}^{\alpha_i+q\Delta\alpha}(n, f_j)_{\Delta t} = \sum_r Z_T(n, f_j + \frac{\alpha_i}{2}) Z_T^*(n, f_j - \frac{\alpha_i}{2}) g(n-r) e^{-j2\pi q\Delta\alpha r T_s} \quad (7)$$

The sum in equation (7) is performed utilizing an N -point FFT, which speeds up the FAM algorithm.

4. Material and methods

In this section, a detailed description of the used database and the proposed method named SCF-convolutional recurrent neural network (CRNN) is provided.

4.1. Database

The proposed method was validated using the 2017 PhysioNet challenge database [32], which consists of 8,528 single-lead ECG recordings obtained from patients in California, United States, using the AliveCor Kardia self-diagnostic device [32]. The ECG signals in this

dataset had varying lengths, ranging from 9 to 60 s, and were sampled at a rate of 300 Hz. Subsequently, a bandpass filtering step was applied as part of the preprocessing procedure. The data was categorized into four classes: atrial fibrillation (class A with 738 samples), normal sinus rhythm (class N with 5,050 samples), other rhythms (class O with 2,456 samples), and noisy recordings (class ~ with 284 samples). A representative waveform for each class is displayed in Fig. 1.

4.2. Proposed method

The ECG signal is transformed into the bi-frequency plane using SCF, and cyclostationary coefficients are obtained. Since the SCF generates multiple coefficients, harnessing the advantages of all these coefficients with a CNN-based approach is feasible. The proposed method is evaluated using the 2017 PhysioNet challenge dataset for ECG classification, as shown in Fig. 2. A detailed description of the ECG representation using SCF and the CRNN architectures used in the proposed approach is provided in the following section.

4.2.1. ECG representation using SCF

In this study, the spectral correlation coefficients of the ECG were estimated using the FAM algorithm, as previously introduced. For data capturing, a Hamming window was employed due to its main lobe width and the amplitude of the side lobes in the frequency domain [26]. The Hamming window is expressed as follows:

$$g(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2\pi n}{N}\right) & 0 \leq n \leq N \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

In this study, N represents the selected window length, set at 512 samples. At the initial stage of the proposed method, the SCF is applied

to a 4096-sample ECG segment. This segment length, approximately half of the total signal length, includes several QRS-T complexes. The 4096-sample length was chosen as it is the longest power of 2 that fits within the available signal data for most records. It is important to note that $8192 \text{ samples} \cdot 2^{13} = 8192$ were not available in all ECG records, necessitating this adjustment.

After applying the SCF to the ECG signals, a bi-frequency plane is generated. The plane bi-frequency plane consists of two frequency axes, with the x-axis representing frequency and the y-axis representing cyclic frequency. To make the bi-frequency plane compatible with the CNN used for classification, we resized it to a 227×227 image.

Fig. 3 provides an example of a bi-frequency plane of spectral correlation coefficients for a typical ECG. The grayscale image initially displays a low dynamic range (Fig. 3, left). To enhance this range, a logarithmic transformation was applied, and the grayscale image was converted to color, allowing for improved visualization and interpretation (Fig. 3, right).

Furthermore, to better understand how the SCF distinguishes different ECG rhythms, it was applied to four different ECG signals belonging to different classes in the 2017 PhysioNet challenge dataset, and their bi-frequency planes were shown in Fig. 4. As observed, the spectral correlation coefficients exhibit distinct patterns in the bi-frequency plane. This indicates that the hidden periodic components in the frequency domain of ECG signals belonging to different classes, such as normal, AF, other rhythm, and noisy signals, have different patterns, as anticipated.

4.2.2. CNN architectures

As before mentioned, SCF generates many coefficients, and DL makes it possible to benefit from maximum potential of these coefficients in separating different classes. To evaluate the effectiveness of using the bi-

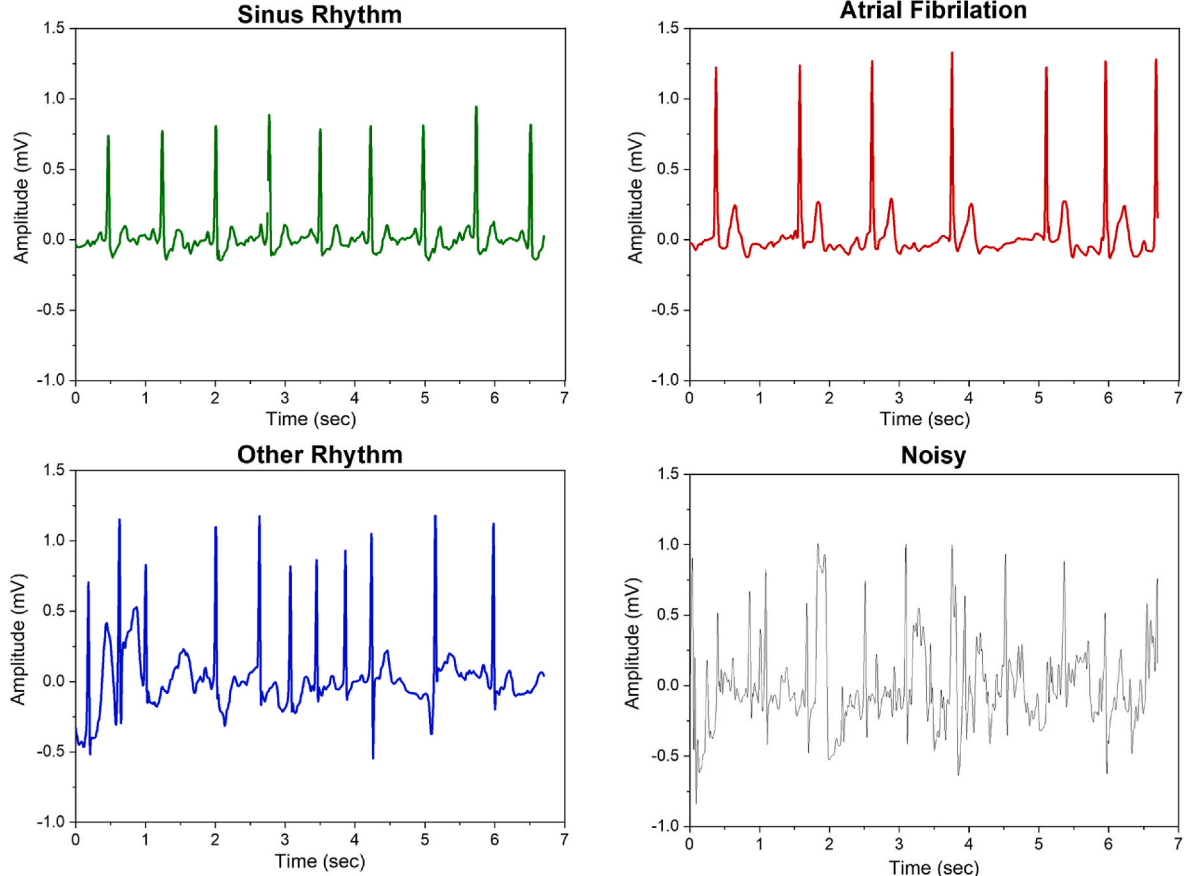


Fig. 1. Typical ECG segments of the four different classes.

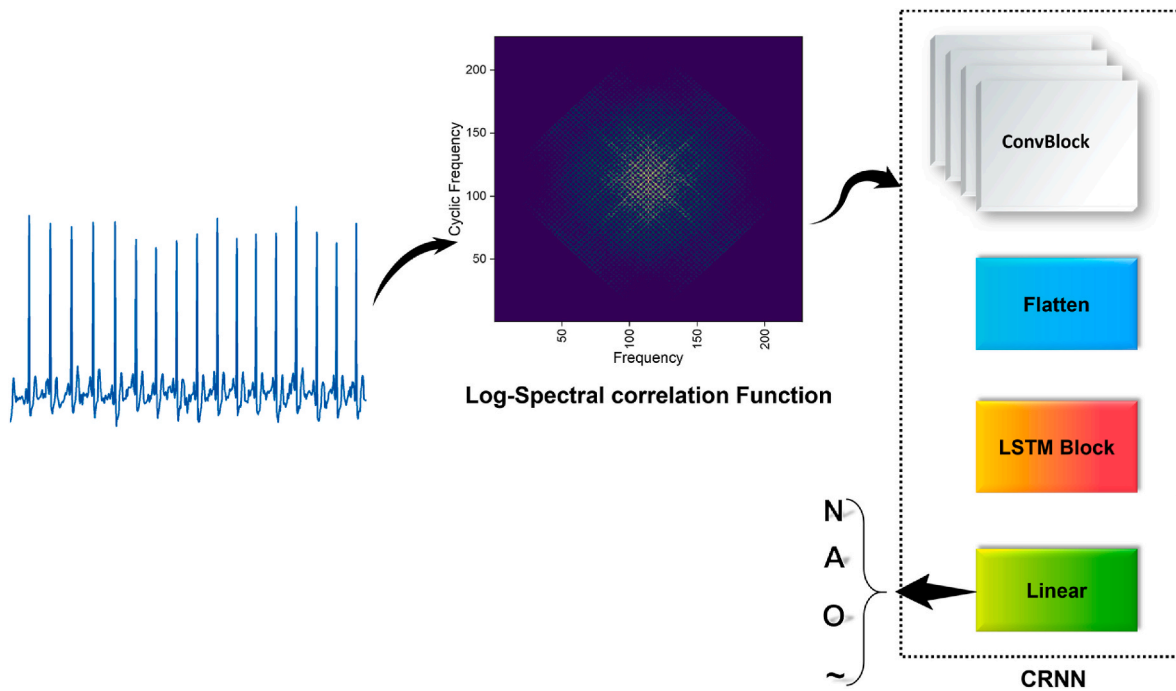


Fig. 2. The block diagram for SCF-CRNN.

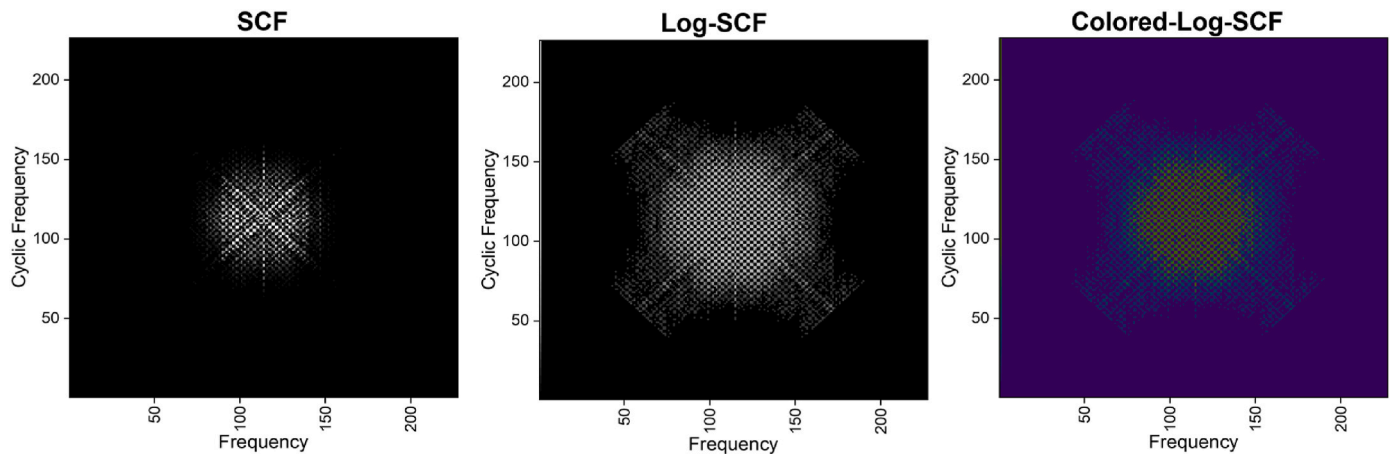


Fig. 3. SCF bi-frequency plane for an ECG signal (left), logarithmic SCF (middle) and colored logarithmic SCF (right).

frequency plane for ECG representation, the estimated SCF coefficients were applied to a recommended DL architecture named CRNN for ECG classification. This CRNN architecture was proposed by Zihlmann et al. [23] and achieved a high rank (5th place) in the 2017 PhysioNet challenge. The CRNN architecture includes four convolutional blocks, one flatten block, one Long Short-Term Memory (LSTM) block, and a standard linear layer with SoftMax, as shown in Fig. 2. Each convolutional block uses a set of 5×5 convolutional filters, batch normalization, ReLU activation, and max-pooling over 2×2 windows. The 3-dimensional feature map is flattened using a flatten block, and then fed into a 3-layer LSTM network. The last layers of the LSTM network provides the feature vector. For more details on the CRNN architecture, please refer to Ref. [23].

4.2.3. Training

To evaluate how well the classification model performs on 2017 PhysioNet challenge dataset, a repeated 10-fold cross-validation is employed. This involves generating training and test sets multiple times,

with each iteration using a different portion of the data for training and testing. The final evaluation is based on the average of the performance measures obtained from all the test sets.

Furthermore the Synthetic Minority Over-sampling Technique (SMOTE) is utilized to address the issue of imbalanced data database. SMOTE works by generating synthetic examples of the minority class by interpolating between examples of the minority class that are already present. This helps to balance the class distribution and improve the performance of classifiers trained on imbalanced datasets [33]. Additionally, to optimize the weight parameters in this study, the “adam” optimizer was employed. The values chosen for the optimization process were an initial learning rate of 0.01 and the batch size was set to 20 [23]. All experiments of this study was executed utilizing Python 3.9 as the programming environment and on a computer with the following specifications: CPU Intel cori7 with RTX 3050, and 16 GB of RAM.

4.2.4. Evaluation

To minimize the impact of chance on the outcomes, all experiments

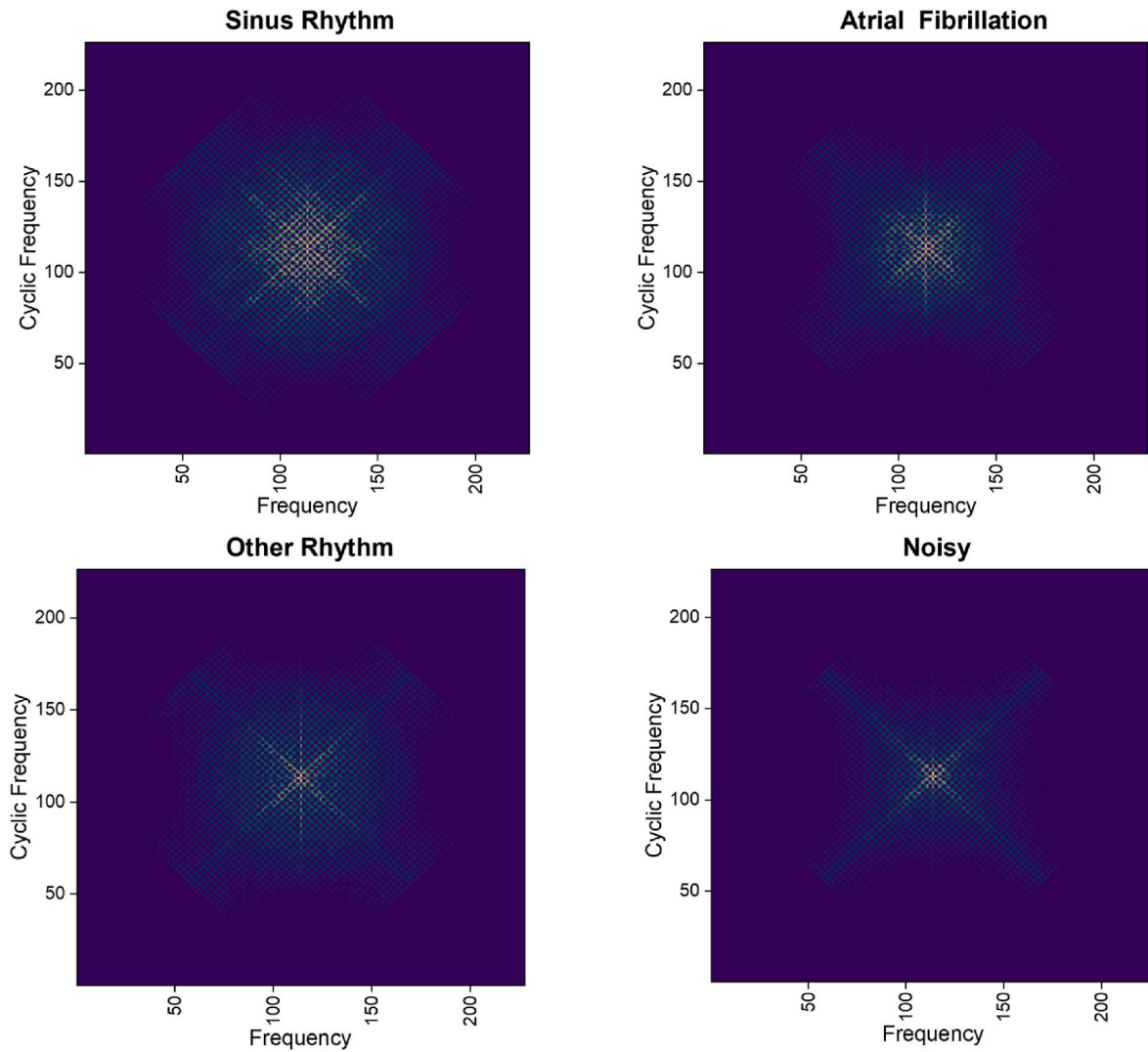


Fig. 4. Typical spectral correlation function of the four different ECG belonging to different classes.

were conducted using a 10-fold cross-validation approach, and the average performance metrics were reported. Several performance metrics were utilized to assess the effectiveness of SCF-CRNN, including Accuracy (Acc), Specificity (Spe), Recall (R), Precision (Pre), and F1-score (F).

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$Sp = \frac{TN}{TN + FP} \quad (10)$$

$$R = \frac{TP}{TP + FN} \quad (11)$$

$$Pre = \frac{TP}{TP + FP} \quad (12)$$

$$F1 - score (F) = \frac{2Pre * R}{Pre + R} \quad (13)$$

where True Positives (TP) refers to the number of signals that are correctly identified as belonging to a particular class, while True Negatives (TN) are the number of signals that are correctly identified as not belonging to that class. False Positives (FP) are signals that are incorrectly identified as belonging to the class, while False Negatives (FN) are

signals that are not identified as belonging to the class, even though they actually do.

5. Results and discussion

5.1. The obtained results of SCF-CRNN

Based on the average results of 10-fold cross-validation experiments, the SCF-CRNN achieved an overall F1-score of 0.891, with accuracies of 0.947, 0.916, and 0.903 for normal rhythms, AF rhythms, and other rhythms, respectively. Fig. 5 shows the distribution of predictions for different classes. However, SCF-CRNN faced some challenges in distinguishing normal and other rhythms classes, with 4% of the normal rhythms were misclassified as other rhythms class, and 7.3% of the other rhythms were misclassified as normal rhythms. Additionally, about 6% of AF rhythms were incorrectly predicted as other rhythms. Nevertheless, SCF-CRNN exhibited relatively small errors in distinguishing normal rhythms from AF rhythms, indicating its effectiveness in differentiating these two classes.

Furthermore, the performance of SCF-CRNN is compared with Zihlmann et al. [23], which applied the CRNN to the normalized logarithmic spectrum of ECG and achieved fifth place in the 2017 Physionet challenge for ECG classification. Table 1 summarizes the comparison in terms of accuracy and F1-score, highlighting the significant impact of

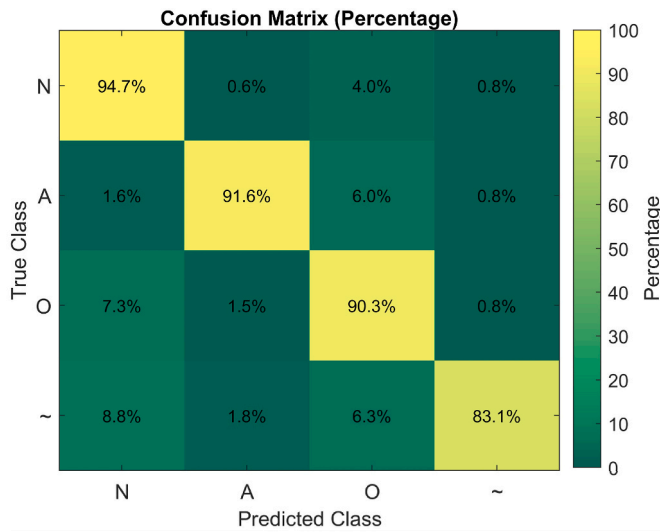


Fig. 5. The confusion matrix for the proposed method.

Table 1

Comparing the performance of SCF-CRNN with CRNN technique in terms of accuracies and F1-scores for ECG classification.

Method		N	A	O	~	overall
Spectrum-CRNN [23]	Acc (%)	90.2	69.1	63	51.1	79.2
	F (%)	87.4	69.9	66.5	54.9	74.6
SCF-CRNN (the proposed method)	Acc (%)	94.7	91.6	90.3	83.1	92.76
	F (%)	95.16	91.04	89.87	80.41	89.12

using SCF coefficients in improving ECG classification performance. The results of Fig. 5 and Table 1 confirm the effectiveness of using SCF for ECG representation.

5.2. Comparison of the obtained results

Table 2 compares the performance of the SCF-CRNN method with other approaches which proposed by different researchers and utilized the 2017 PhysioNet challenge dataset. The evaluation measures F_A , F_N , F_O , $F_{\sim}$, and F_{mean} , which correspond to the F1-score for the classes A, N, O, and $\sim$, were obtained by averaging the results of 10-fold cross-validation for the proposed methods. The obtained scores were then compared with those reported by the selected methods of the 2017

Table 2

Comparison of the performance of SCF-CRNN with best entries of the 2017 PhysioNet challenge and other state-of-the-art methods.

Method	F_N (%)	F_A (%)	F_O (%)	$F_{\sim}$ (%)	F_{mean} (%)
Cascade classifier [34]	79	91	77	–	83
Random Forest [7]	79	91	76	61	82
Hand-craft features & CNN [13]	96	84	86	72	88
CRNN [23]	87	69	63	55	75
Deep CNN [34]	92	82	75	–	83
Decision Tree Ensemble [35]	93	86	79	–	86
ENCASE [36]	92	87	81	82	85
Hybrid Network [37]	93	88	74	–	85
Hand-craft features & random forest [9]	76	90	73	57	74
CNN-SVM [38]	93	79	79	–	84
Hand-craft features & XGB [39]	85	83	75	–	75
SCF-CRNN (the proposed method)	95.16	91.04	89.87	80.41	89.12

PhysioNet challenge [8, 9, 14, 27, 46, 47, 48, 49], as well as recent studies conducted on the 2017 PhysioNet challenge dataset. The results show that the proposed method outperforms state-of-the-art techniques for AF detection in terms of F_A , F_O , and F_{mean} . Additionally, although the obtained F1-score for normal and noisy classes are not the highest, the results obtained with the SCF-CRNN method are very close to those of the other best methods.

Moreover, the results were compared with several state-of-the-art methods in terms of accuracy, specificity, recall, precision, and F1-score, as shown in Table 3. The proposed method achieved 92.76% accuracy, 96.93% specificity, and 89.92% recall on the 2017 PhysioNet challenge database, demonstrating superior performance in ECG classification compared to other approaches. The use of SCF to represent ECG signals in the frequency domain enhances the method's effectiveness in distinguishing between various ECG classes.

While SCF-CRNN has demonstrated effective performance compared to top methods in the 2017 PhysioNet challenge, it does have two weaknesses. Firstly, because of ECG segmentation, since the dataset does not provide information on where certain rhythms, such as AF, occur in each ECG signal, making it is possible that the selected ECG segment may not contain important information related to the dominant rhythm. To address this, a segment length of 4096 samples is selected, roughly half the length of the entire signal, to increase the likelihood that each segment contains important signal patterns. Additionally, SCF-CRNN utilizes SCF to reveal hidden patterns in the signal, which I believe that these patterns are inherently present in the signal throughout its duration. Secondly, the proposed method suffers from the high processing time required for SCF and CRNN computation. In future work, I plan to conduct an in-depth study of SCF and benefit bi-frequency plane symmetry to significantly decrease the processing time.

6. Conclusion

In this study, the SCF was utilized to estimate the cyclic correlation coefficients and investigate the relationship between the QRS-T complex of ECG signals. The effectiveness of this approach was demonstrated by employing the deep learning technique for ECG classification on the public databases of the 2017 Physionet challenge. The SCF provides a bi-frequency plane that transforms the ECG signals into images. To fully leverage the estimated spectral correlation information, 2D convolutional layers were used in the initial layers of the DL architecture. The results supported the cyclo-stationary hypothesis of ECG signals, as the proposed method outperformed other state-of-the-art methods used in the PhysioNet 2017 challenge and recent publications. These findings suggest that the proposed method has the potential to be employed for analyzing various biomedical signals, including EEG and phonocardiogram signals.

Table 3

Average results of 10-fold cross-validation of SCF-CRNN comparing with other methods.

Method	Acc (%)	Spe (%)	R (%)	Pre (%)	F (%)
Hand-craft features [6]	–	–	70.5	73.8	70.6
Deep learning [18]	–	–	71.3	77.9	74.2
Hand-craft features & random forest [9]	83.3	92	72	78.3	74.46
GoogleNet ensemble with AlexNet [40]	92.50	–	–	–	–
Multi-branching ResNet [41]	–	–	–	–	88.65
Dynamic transfer learning & VGG16 [42]	91.4	–	–	–	87.30
SCF-CRNN (the proposed method)	92.76	96.93	89.92	88.36	89.12

Author contributions

The manuscript was written by Sara Mihandoost. S.M. conceptualized the modeling and computational sections and wrote the manuscript.

Declaration of competing interest

All other authors declare that they have no conflicts of interest regarding the publication of this paper.

Data availability

Data will be made available on request.

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